

Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Demographic Faces and Deepfake Image Detection

Project Supervisor

Supervisor



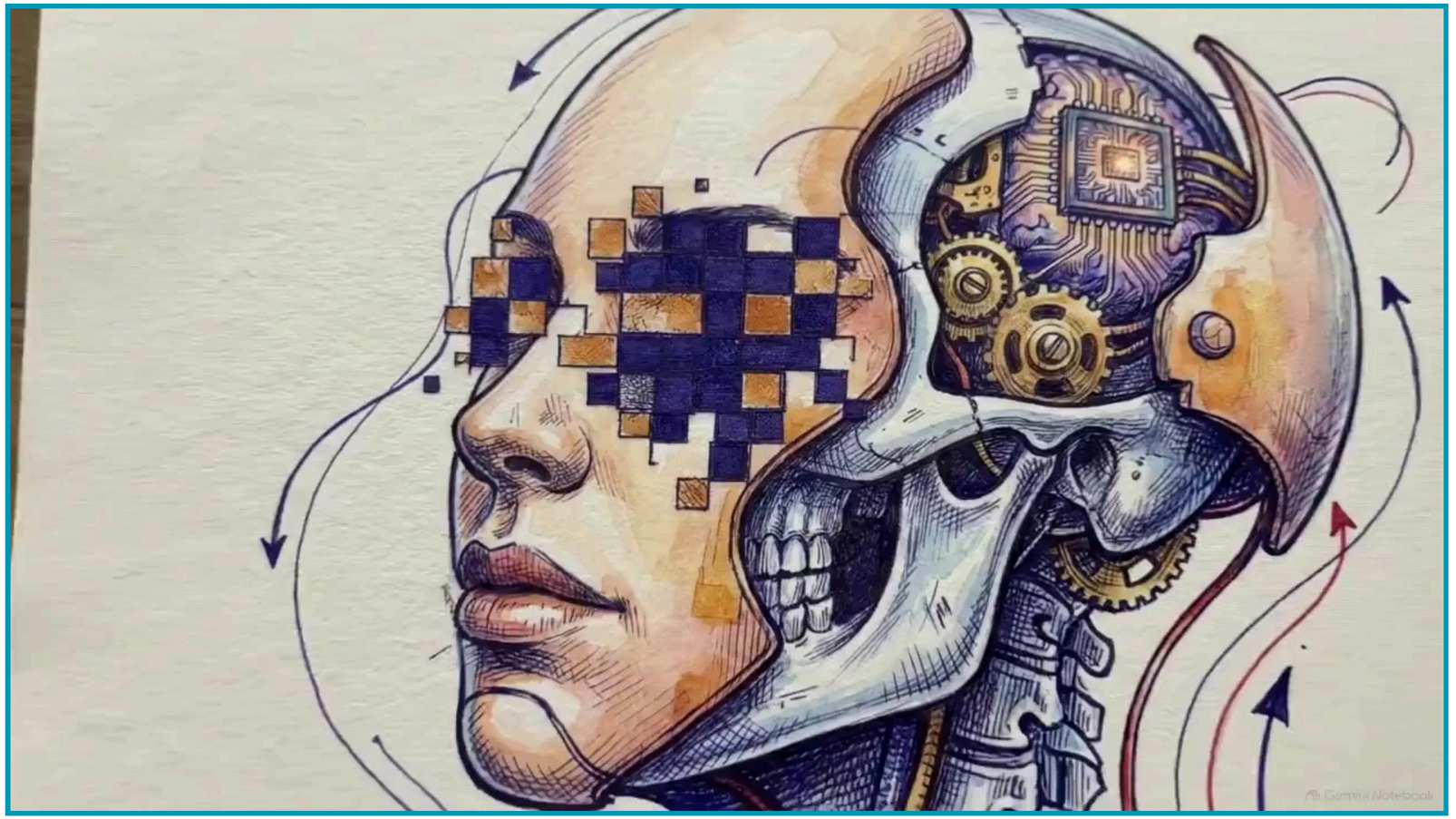
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FACEO Demonstration



The Problem

The rapid advancement of computer vision and deep learning has made facial analysis widely accessible, yet this capability remains fragmented across disconnected, purpose-built tools. Simultaneously, the proliferation of AI-generated synthetic media commonly known as deepfakes has introduced unprecedented threats to digital trust, identity verification, and personal safety. Traditional systems address only isolated aspects of these challenges, forcing practitioners and researchers to integrate multiple, incompatible tools with inconsistent accuracy and no unified interaction surface.

How Faceo Addresses the problem

FACEO (Facial Analysis & Comprehensive Evaluation Open-platform) is a unified, open, real-time facial intelligence system that runs across the full stack from client-side browser inference (TensorFlow.js / face-api.js) to server-side microservices (FastAPI, DeepFace, OpenCV) solving all five problem domains simultaneously through a single, coherent user interface.

Users and Stakeholders

- Primary end users: General public, students
- Professional/institutional users: HR, law enforcement, child welfare
- Secondary stakeholders: Researchers, developers, regulators, ethics bodies

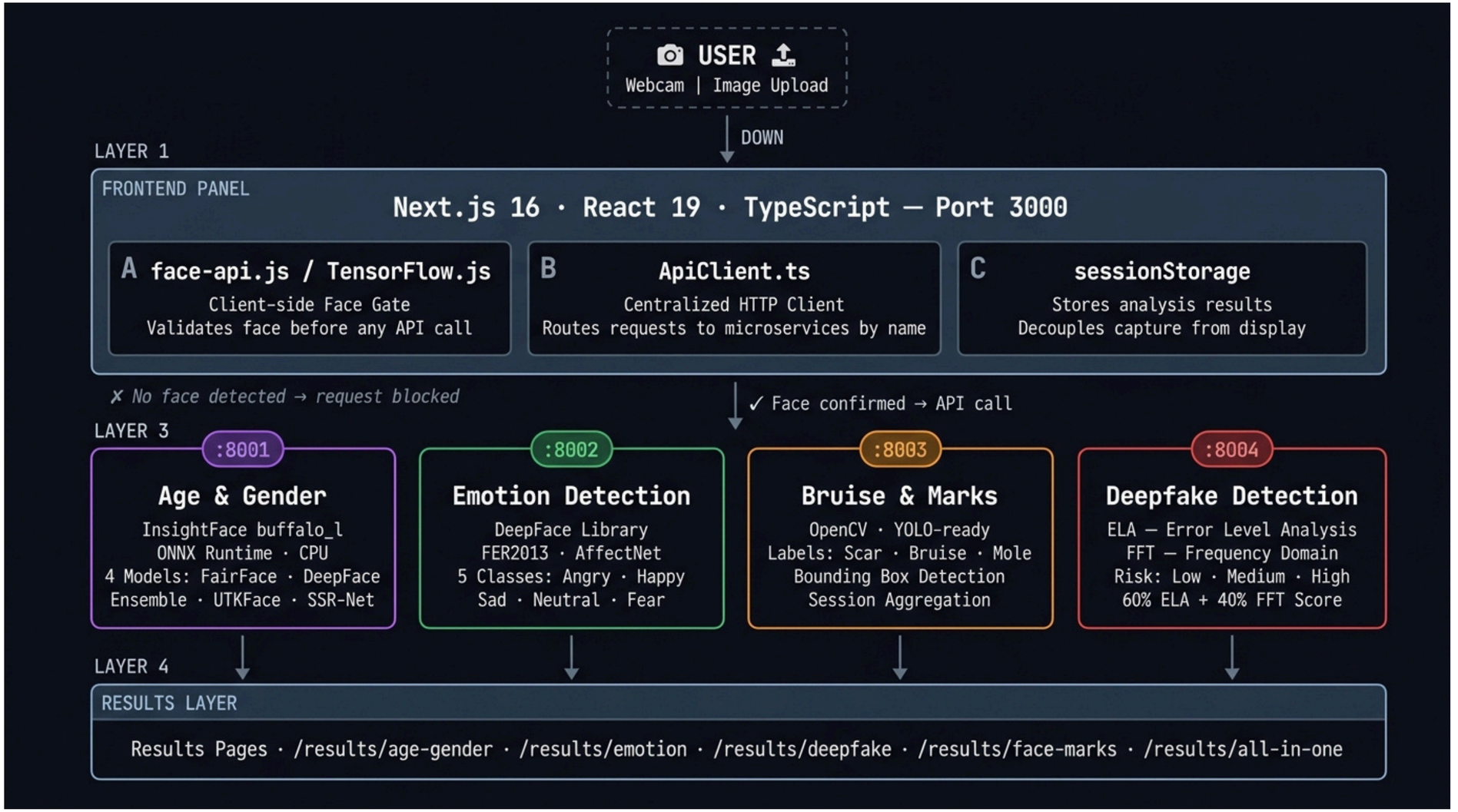
Impact on users and society

Impact Area	*Beneficiary*	*Nature of Benefit*
Deepfake detection	General public, media, law enforcement	Protects information integrity and evidence reliability
Bruise/mark detection	Children, welfare workers, investigators	Strengthens safeguarding and forensic processes
Emotion recognition	Educators, mental health practitioners, developers	Enables empathetic, adaptive human-computer interaction
Demographic estimation	Researchers, ethicists, product teams	Promotes fairness and bias accountability in AI systems
Open-source platform	Academic and developer community	Accelerates responsible AI research and education

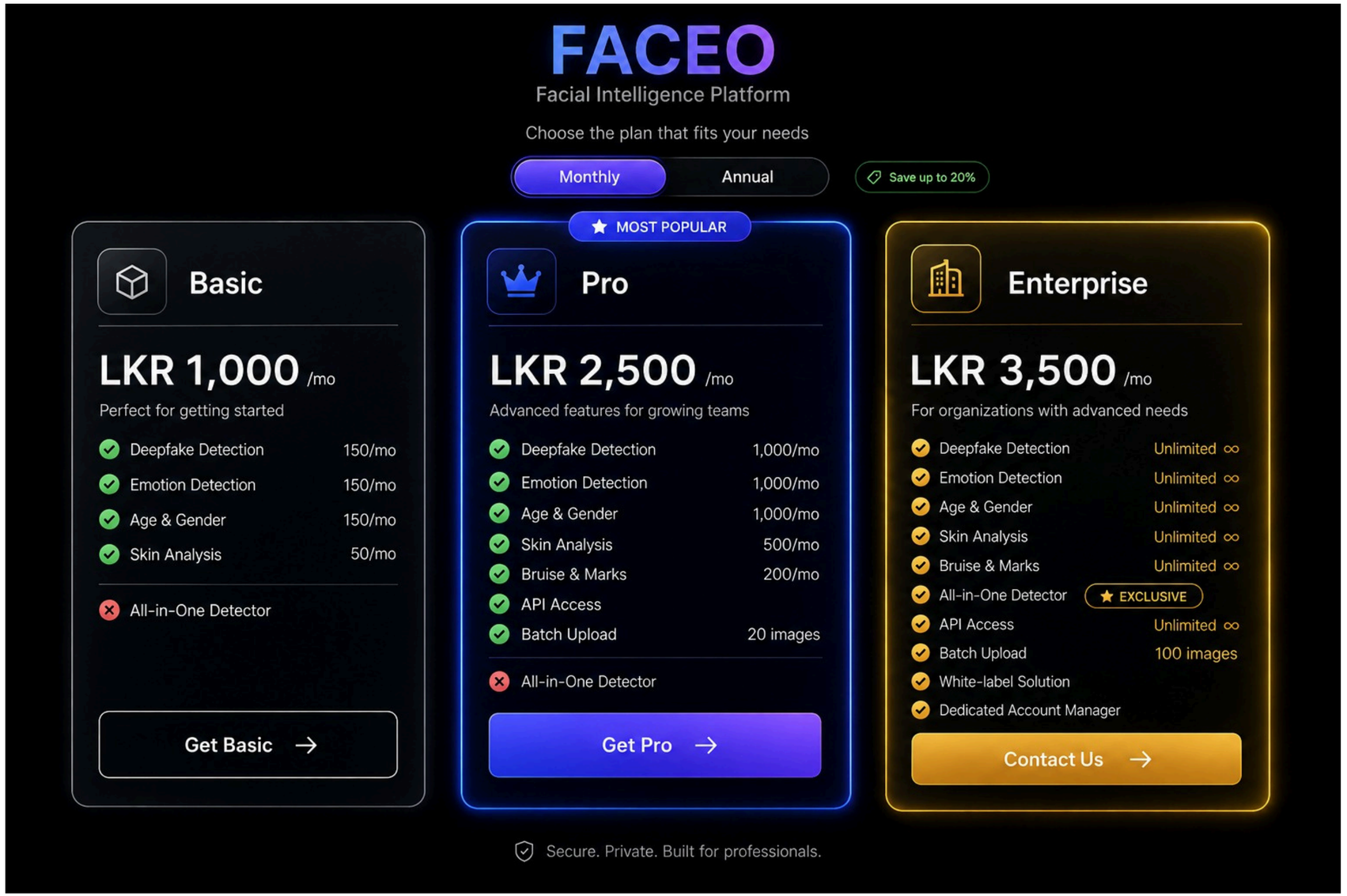
Comments for the PP1

Student ID	*Comment*	*How we addressed*
IT22541970	* Moderate idea about component. * Do a model comparison with 2-3 models on model performance.	Implemented and fine-tuned two emotion recognition models for performance comparison. Additionally, the SSD MobileNet model was fine-tuned and enhanced to improve emotion recognition performance under low-light conditions.
IT22254252	* Detects the age and the gender with a real-capture of a person, and with a few seconds, generate a report with the details, and the model needs further fine-tuning and optimization, as the results were incorrect, needs major improvement. * Moderate idea, but needs to do a model comparison with 2-3 models on model performance, have trained only 1 model.	The engine was replaced from DeepFace to InsightFace buffalo_l (ONNX), reducing age error from ±5–8 years to ±2–3 years. A custom age calibration curve was added to correct bias on South Asian faces. Four model variants are now selectable FairFace, DeepFace Ensemble, UTKFace ResNet, and SSR-Net/MobileNet with live webcam capture generating a report within seconds.
IT22278876	* Component has been changed to 'Suspicious facial detection' addressing panel comments. * Lacks idea about component, need discussion with supervisor. * Do a model comparison with 2-3 models on model performance, have trained only 1 model.	The component scope was clarified to detect bruises, scars, and moles using bounding-box detection. The full API and UI pipeline is built. YOLO model integration is planned as the next development step.
IT22276728	* Detecs the real and fake images when uploaded. * Do a model comparison with 2-3 models on model performance, have trained only 1 model. * Moderate idea	Implemented and evaluated three distinct deepfake detection models (ResNet-50 CNN, Vision Transformer, and Swin Transformer) for a comprehensive performance comparison. Additionally, a fully functional web application was developed for detection, and cross-dataset evaluation was conducted to assess model robustness against heavily compressed and unseen artifacts.

Sentinel System Architecture



Commercialization



Demo

Q & A

Thank You